Fuel Level Control Valve – Maintenance Practices

1. Removal/Installation

   A. Remove Fuel Level Control Valve – in Fuselage Tank

      1) Ensure battery and/or DC power (refer to 12-00-00, Servicing – General Data, Para 2.) is disconnected, crossfeed valve closed, and wing fuel tank switches (if fitted) are OFF.

      2) Drain applicable forward or aft fuselage fuel tanks. (Refer to 12-10-00, Fuel System – General Data).

      3) Remove access panel from underside of fuselage. (Refer to 12-00-00, Servicing – General Data).

      4) Remove applicable sump plate. (Refer to 28-10-21, Fuel Cell Slump Plate – Maintenance Practices).

      5) Remove level control valve retaining bolts and washers. Remove valve and gasket from end of tube assembly.

   B. Install Fuel Level Control Valve in Fuselage Tank

      1) Position valve with new gasket on end of tube assembly.

      2) Install four retaining bolts and washers. Lockwire bolts.

      3) When installing valve to sump plates of cells 4 or 5, ensure blanking cap is fitted to tee at base of tube assembly.

      4) Install sump plate. (Refer to 28-10-21, Fuel Cell Slump Plate – Maintenance Practices).

      5) Check for fuel leaks.

      6) Install access panel. (Refer to 12-00-00, Servicing – General Data).

   C. Remove Fuel Level Control Valve in Wing Tank (Mod S.O.O. 6247)

      1) Connect battery and/or DC power (refer to 12-00-00, Servicing – General Data, Para 2.) to aircraft.

      2) Drain applicable wing fuel tank (if necessary). (Refer to 12-10-00, Fuel System – General Data).

      3) Remove wing tip.

      4) Disconnect external fuel line from level control valve elbow at tank outboard face.

      5) Remove tank filler cap and use as access to level control valve.

      6) Support level control valve inside tank and remove retaining bolts and washers from outside tank.

      7) Remove level control valve and attached elbow assembly through filler cap opening. Ensure tank to elbow gasket is removed with elbow assembly.

      8) Cut lockwire and remove elbow to level control valve retaining bolts and washers. Remove level control valve from elbow. Retain elbow.

   D. Install Fuel Level Control Valve – in Wing Tank (Mod S.O.O. 6247)

      1) Position valve with new gasket on elbow.

      2) Install four retaining bolts and washers. Lockwire bolts.

3) Position four retaining bolts and washers to protrude into tank, and working through filler cap, position new gasket over bolts inside tank.

4) Position level control valve and attached elbow inside tank and secure with retaining bolts.

5) Connect external fuel line to level control valve elbow at tank outboard face.

6) Install wing tip.

2. Adjustment/Test

A. Bench Test

Equipment required:

- Controlled air supply 0 to 125 psi
- Controlled test fluid supply (dry cleaning solvent P–D–680) 0 to 125 psi pressure and 0 to 20 gpm flow
- Suitable test fixture

1) Proof Pressure Test Valve

a) Connect air line to valve inlet port.

b) Mount valve upright in suitable test fixture, submerged in test fluid.

c) Apply air to inlet port at increasing pressures up to 125 psi. Check no external leakage occurs.

2) Leak Test Valve

a) With valve in inverted position, apply 60 psi test fluid pressure to inlet port.

b) Check for internal leakage (maximum allowable 31cc per minute).

3) Test Sensing Level of Valve

a) Mount valve upright in suitable test fixture.

b) Apply test fluid at a flow rate of one gpm to valve inlet port.

c) Check valve closes (no further fluid flow) when sensing level of fluid is 3 1/4 ± 3/16 inches above the base of the mounting flange.

4) Pressure Drop Test Valve

a) Mount valve upright on a one-inch diameter calibrated pressure tap.

b) Connect fluid source to inlet and apply test fluid at a flow rate of 20 gpm through valve.

c) Check pressure drop does not exceed 2 psi.

d) Remove valve from test fixture.